Contribution of sound simulations by physical model and machine learning models for the prediction of spectral centroid trumpets sounds: influence of the leapipe bore

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1. ABSTRACT

This work consists in studying the contribution of sound simulations by physical model and machine learning (ML) models for the prediction of trumpet's sounds spectral centroid. A training set of 1000 virtual instruments is constituted by varying the internal geometry (the leadpipe) of the bore of a trumpet. These instruments are then simulated using a physical model of the instrument-musician couple, with different dynamics by varying the mouth pressure. For each instrument, two descriptors are calculated from the simulations: the EFP (equivalent fundamental pitch, representing the intonation of the notes created), and the spectral centroid of the sounds (representing the brightness). From the spectral centroid with different dynamics, two spectral variables are calculated: the average spectral centroid and the spectral enrichment rate. A supervised learning is then performed on the training set, modeling each variable according to the geometric variables of the leadpipe. Different maps are then proposed to show how the two spectral variables evolve as a function of the geometry of the leadpipe. Intonation constraints of the instrument are added, using the EFP descriptor. The maps allow visualization of typical leadpipe profiles, corresponding to given levels of the spectral variables. Different geometries are proposed, which could be tested by manufacturing real prototypes and confirm the validity of the approach.

2. INTRODUCTION

An interesting way to assist the design of musical instruments is to carry out sound simulations by physical modelling[1]. It constitutes a very interesting approach because they allow, by working on a virtual prototype, the exploration of the design space by the creation of a large number of virtual instruments. The main interest of these simulations is that the sound result is driven by the causes that create the sound, as for a real instrument: if the physical model used is accurate enough to generate simulations in agreement with the real behavior (such as it is perceived by the musician), then the simulations can constitute a predictive tool for the development of the instrument (virtual prototyping)[2].

In this study, we use a classical elementary model of a brass instrument under playing conditions[3], that couples the vibrating lips (modeled as a one-degree-of-freedom outward-striking valve) to the air column of the brass instrument (represented by its input impedance Z_{in}). For a set of virtual instruments (represented by their input impedances, or their internal geometry), different regimes of the trumpet can be simulated. From this database of synthesized sounds, different criteria can be considered for the optimization of the bore of the trumpet: the Equivalent Fundamental Pitch (*EFP*), which may represent the global intonation of the instrument, the minimum blowing pressure (*Pth*), which may represent the ease of playing of the note, the spectral centroid (*SC*) of the sounds, representing the brightness. Although simulations are computationally intensive, it is possible to predict performance criteria across the entire design space using machine learning tools.

Indeed, machine learning (ML) has become an essential approach for the modeling of systems. The principle of our approach is to train a Machine Learning (ML) model with a set of simulated sounds (training set), with characteristics of the trumpet (input impedance or bore geometry) as inputs and performance criteria (*EFP*, P_{th} or *SC*) as outputs. These studies focus on optimizing the trumpet, in particular the leadpipe of the trumpet:

- A biobjective optimization of the input impedance according to (*EFP*, P_{th}) is presented in Ref. 4., leading to the definition of an "optimal" input impedance. A bore reconstruction procedure is finally applied in order to obtain the geometry of the leadpipe, corresponding to the optimal input impedance. Details of the approach can be found in Ref. 1 and 4, for which a leadpipe prototype was manufactured by *Yamaha Corporation of Japan.* A comparison of the performances of the virtual and real instruments is presented in Ref. 11,
- Another biobjective optimization of the leadpipe according to (*EFP*, P_{th}) is presented in Ref. 10. In this paper, instead of using the input impedance as inputs of the ML models, the objective was to use directly the geometry of the bore of the instrument (the leadpipe). This approach has the great advantage of not requiring the bore reconstruction phase, which requires to solve a complex inverse problem.

This paper is a continuation of this work. The objective is to consider now criteria related to the timber of the instrument for the optimization of the leadpipe geometry, in particular the spectral centroid of the sounds.

Section 2 presents the background on the project, mainly the procedure for the sound simulations and the parameterization of the trumpet leadpipe geometry. Section 3 presents the methods used for the simulations and the machine learning models. In section 4, the first results are presented. They are presented in the form of maps, useful for visualizing the influence of the leadpipe geometry on the average spectral centroid and the spectral enrichment rate.

3. BACKGROUND: SOUND SIMULATIONS AND LEADPIPE PARAMETERIZATION

A. SOUND SIMULATIONS

This study uses a classical elementary model of a brass instrument under playing conditions, described in Ref. 2. The vibrating lips are modeled as a one-degree-of-freedom (1-DOF) outward-striking valve, non-linearly coupled to the air column of the brass instrument. From the input impedance Z_{in} of the instrument (calculated or measured), several regimes of the trumpet can be simulated using different virtual musicians (M virtual musicians). The parameters used for the simulations are described in Ref. 1 and Ref. 4. A set of M = 6 virtual musicians was defined to play each instrument and five regimes of the instrument were considered (regime 2 to 6), corresponding to the notes Bb3, F4, Bb4, D5 and F5 (concert-pitch) (Fig. 1).

Figure 1. Notes of a Bb trumpet (concert pitch) corresponding to the five regimes simulated (2 to 6).

To characterize each sound simulated, the fundamental frequencies were estimated using the YIN algorithm[5]. For each regime simulated, the average playing frequency $\overline{f_n}$ for all of the *M* virtual musicians was computed. To characterize the intonation of each regime, the Equivalent Fundamental Pitch (*EFP* – Eq. (1)) is calculated. It represents the deviation in cent of the average playing frequency $\overline{f_n}$ from a reference frequency $\overline{f_R}$, according to natural intervals (the choice was made to consider harmonic intervals instead of equal temperament pitches). The reference frequency was chosen arbitrarily according to the common tuning note of the instrument, the regime 4 (Bb4) (with $f_R = f_4/4$, the *EFP* of the regime 4 is then necessarily equal to "0").

$$\overline{EFP_n} = 1200.log_2(\frac{\overline{f_n}}{n.\overline{f_R}}) \quad (1)$$

To characterize the global intonation of each virtual instrument, the global average EFP was computed. It corresponds simply to the average value of the absolute values of the EFP across all the regimes (2 to 6) (Eq. (2)).

$$EFP = \frac{1}{5} \cdot \sum_{k=2}^{6} |\overline{EFP_k}| \tag{2}$$

Detailed results concerning the intonation of simulated sounds can be found in Ref. 10 and Ref. 11.

B. PARAMETERIZATION OF THE LEADPIPE GEOMETRY

Instead of varying the whole bore of the trumpet, it was decided to focus only on an important part of the bore: the leadpipe. The leadpipe of a trumpet, located after the mouthpiece and before the tuning slide, plays an important role in the response of the instrument, particularly in the intonation. Its shape is evolutive, generally divergent. At the end of the process, only the leadpipe will need to be manufactured, resulting in a cost-effective and faster solution to test the prototypes. The rest of the instrument corresponds to a *Yamaha* trumpet model for which accurate impedance measurements are available.

Six optimization variables were defined in the leadpipe, 5 radii and one length (Fig. 2). The leadpipe is considered as a series of four truncated cones. The design variables are then defined by vector X=(R_1, R_2, R_3, R_4, R_5, L). Only one truncated cone, the fourth, has a variable length, in order to be able to change the total length of the leadpipe.

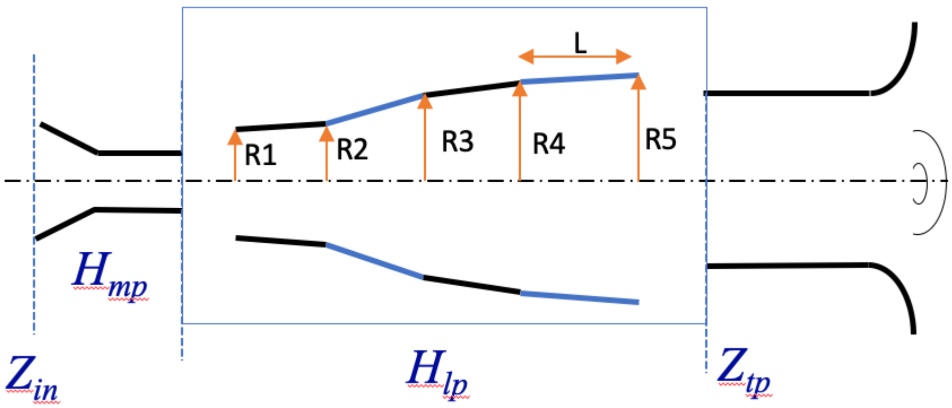

Figure 2. Overview of the bore of the trumpet (from the mouthpiece to the bell), with the definition of the design variables. The optimized part corresponds to the leadpipe, with the radii R_1 to R_5 and the length L.

To obtain the current input impedance of the trumpet, a transmission line model of the impedance was used[7]. The input impedance of the instrument is a result of a calculation that involves three terms:

- The transfer matrix of the mouthpiece (*Yamaha* model), denoted as H_{mp}
- The transfer matrix of the four sections of the leadpipe, denoted as H_{lp}
- The input impedance Z_{tp} of the rest of the trumpet at the entrance of the tuning slide (a *Yamaha* model), measured with a sensor developed and commercialized by the Center of Technology Transfer of Le Mans (CTTM)[8].

Finally, the input impedance Z_{in} of the whole instrument is calculated as follows (Eq. (3)):

$$H = H_{mp} . H_{lp}$$

$$[P_{in} \; U_{in}] = H . [Z_{tp} \; 1] \tag{3}$$

$$Z_{in} = \frac{P_{in}}{U_{in}}$$

4. MATERIAL AND METHODS

A. GENERATION OF THE DATABASE OF SOUNDS

Sound simulations by physical modelling were used to create a database of sounds. A set of 1000 virtual trumpets was created by a sampling of the leadpipe geometry, according to the design variables X=(R1, R2, R3, R4, R5, L). Realistic values for the lower and upper limits for Ri and L were defined. 1000 geometries were randomly defined, under the constraint that the radius always increases along the length of the leadpipe (divergent shape). This constraint is necessary to be able to manufacture the leadpipe with a mandrel in only one part. From the geometries, the input impedance of the instrument was calculated (Eq. (3)) and the sounds were simulated.

From the sounds of regimes 2 to 6, the global average EFP (Eq. (2)) was calculated. To characterize the timber of the sounds, the spectral centroid SC was considered. It corresponds to the power-weighted average spectral frequency (Eq. (4)).

$$SC = \frac{\int_0^N f.|S(F)|^2 df}{\int_0^N |S(F)|^2\, df} \tag{4}$$

For the spectral centroid, calculations were restricted to regime 2 (Bb2). This choice is motivated by the greater variability observed in the target spectral descriptors within this regime. Other regimes could be considered in further studies. To get a dimensionless descriptor, the normalized spectral centroid *NSC* was calculated by dividing by the playing frequency (Eq. 5).

$$NSC = \frac{SC}{f_2} \tag{5}$$

To account for the shift in the spectral centroid as a function of the mouth pressure (spectral enrichment), different mouth pressure values were considered for the simulations. Given the threshold mouth pressure P_{th} (obtained by linear stability analysis - LSA), simulations were done for $P_m = \alpha.P_{th}$, with $\alpha \in \{1.5, 2, 2.5, 3, 4.5, 6, 9, 10\}$. In total, a set of $N_\alpha = 8$ distinct mouth pressures were used for the simulations. This approach enables a more detailed characterization of the spectral response to blowing pressure, with the specific aim of analyzing brightness evolution for different instrument designs.

The first descriptor used to characterize trumpet *k* computes the average normalized spectral centroid across the different pressure coefficients, for all the M virtual musician (Eq.6).

$$\overline{NSC_k} = \frac{1}{N_\alpha}.\sum_{i=1}^{N_\alpha} \frac{1}{n_{mus}}.\sum_{m=1}^{n_{mus}} NSC_k^m(\alpha_i) \tag{6}$$

The second descriptor represents the evolution of the *NSC* with respect to the pressure in the mouth (spectral enrichment to an increase of the blowing pressure). The resulting averaged *NSC* values exhibit a relatively linear dependence on the blowing parameter α, as shown in Fig. 3 for five instruments from the dataset (randomly chosen).

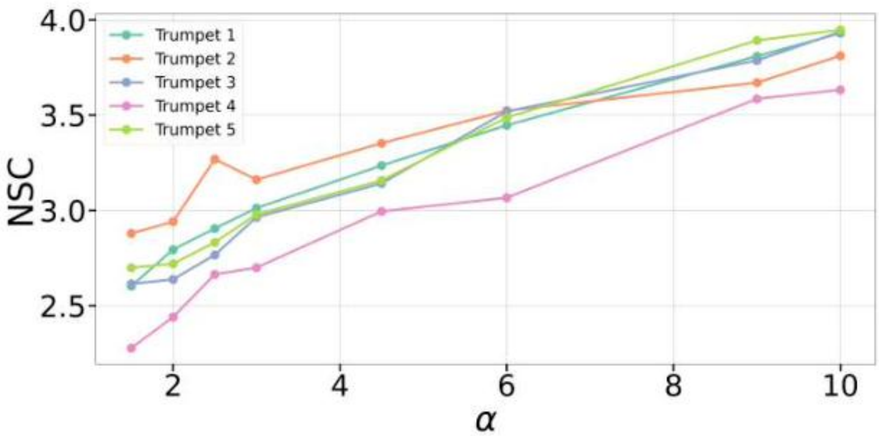

Figure 3. Average NSC with respect to the values of α for a set of 5 trumpets of the dataset

In order to extract this information, a linear regression is fitted to the *NSC* values with respect to the blowing pressure coefficient α, characterizing how average *NSC* varies with pressure (Eq. 7).

$$NSC_k(\alpha) = m_k.\alpha_i + b_k \tag{7}$$

Hence, m_k denotes the slope of the linear fit. While it is well known that spectral centroid increases with larger blowing pressure, m_k captures how rapidly this enrichment occurs. A higher value of m_k indicates that the instrument is more sensitive to changes in blowing pressure. In summary, two descriptors are used to represent the timbre of the trumpet (calculated on the note Bb2):

- $\overline{NSC_k}$ represents the average normalized spectral centroid of the instrument across different blowing pressures P_m,
- m_k denotes the slope of the spectral enrichment, representing the rate of increase of the normalized spectral centroid with respect to P_m.

B. SUPERVIZED LEARNING

ML models were fitted to the set of sounds (training set), with the characterization of the geometry of the leadpipe X=(R_1, R_2, R_3, R_4, R_5, L) as input and the 2 descriptors $\overline{NSC_k}$ and m_k as outputs (in the following, subscript *k* is omitted for the sake of simplicity). The training set counts 1*8*973*6 = 46 704 sounds:1 regime, 8 pressure coefficients, 973 instruments (27 among the 1000 samples failed in the simulations), 6 virtual musicians. The simulation process took approximately 70 hours on a laboratory PC with an Intel Core i7-3770 processor and 8GB RAM.

To ensure robust model training and fair evaluation, the dataset was randomly partitioned into three subsets (training, validation, and test) using the *train_test_split* function from Python's *scikit-learn* library. Data was divided as follows:

– Training set (70%): used to fit the models.

– Validation set (10%): used for hyperparameter tuning and model selection.

– Test set (20%): reserved for final performance evaluation.

The data was partitioned to ensure that the test set remained entirely isolated from the model selection and hyperparameter tuning processes. Model training was conducted using 10-fold cross-validation on the training set to assess the predictive performance of each candidate model while tuning hyperparameters where applicable.

The selection of the most appropriate model for each target variable was guided by the minimization of the Root Mean Squared Error (RMSE), which served as the primary evaluation metric (Eq. 8).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad (8)$$

Hence, the following supervised regression models were evaluated:

- Random Forest Regressor (RF)
- Decision Tree Regressor (CART, default settings)
- Decision Tree Regressor (with squared_error criterion)
- Bagging Regressor
- AdaBoost Regressor
- XGBoost Regressor

In order to enhance model accuracy and prevent overfitting, a thorough hyperparameter tuning was carried out. The hyperparameters considered during tuning were:

- n_estimators (number of trees): 100 to 500
- max_depth (maximum tree depth): 4 to 10
- min_samples_split (minimum number of samples required to split an internal node): 2 to 10
- learning_rate (for boosting models): 0.01 to 0.3
- max_samples (for Bagging): 0.6 to 1.0

The best-performing model for each descriptor was selected based on the lowest cross-validated RMSE obtained during training. This approach reduces the risk of overfitting and provides a robust estimate of the model's generalization capability. Once the model with the lowest RMSE across the cross-validation folds was identified, it was retrained on the combined training and validation datasets. This retraining step fits the model parameters using all available labeled data (excluding the test set), thereby leveraging a larger dataset to improve the model's learning and predictive capability. Finally, the model's predictive performance was evaluated on the independent test set, which was kept completely unseen during both the training and model selection phases. This evaluation provides an unbiased estimate of the model's ability to generalize to new, unseen data and confirms the reliability of the selected model for practical applications.

C. VISUALIZATION OF THE RESULTS

To give a graphical interpretation of the ML models, a sampling of the whole design space was conducted, the responses being calculated for all the samples with the ML models. First, biplots of all the samples according to the two descriptors can be provided. Second, in order to show the distribution of the design variables (R_1, R_2, R_3, R_4, R_5, L) according to different values of the descriptors, violin plots are proposed. Violin plots[12] combine a box plot with a kernel density estimation to visualize both the distribution and the probability density of a variable, making them particularly useful for comparing distributions across different categories or conditions. This method enables the identification of characteristic geometric patterns associated with specific descriptor values, thereby uncovering potential trends linked to particular acoustic behaviors.

Fig. 4 shows for example the distribution of the design variables for two cases concerning the EFP descriptor: valid (green) (EFP<10 cents); non-valid (red: EFP>22 cents).

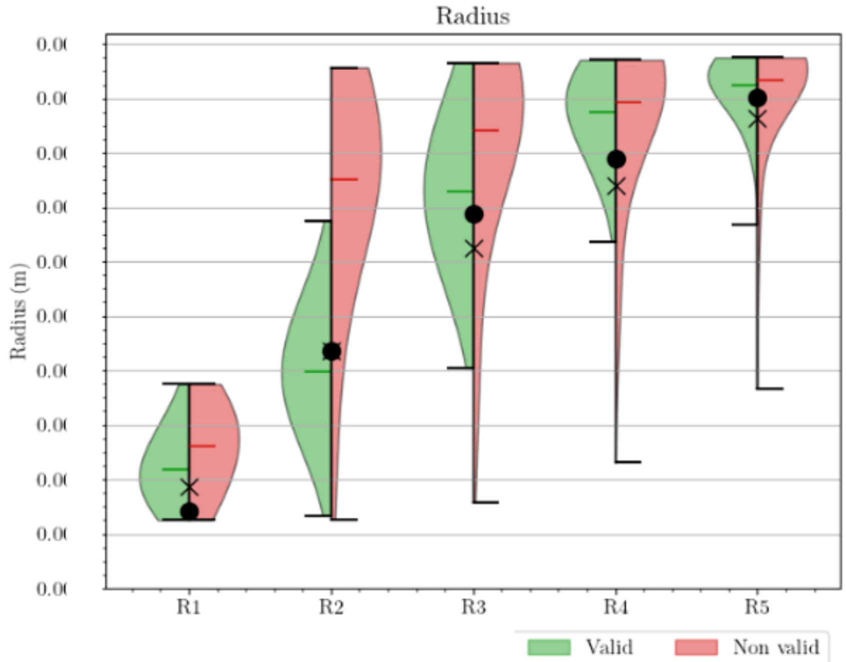

Figure 4. Distribution (violin plots) of the design variables for two cases: intonation valid (green - EFP<10 cents); non-valid (red: EFP>22 cents). For confidentiality reasons, radii values have been removed

This graph provides guidance on how design variables should vary for a given performance level. Here the opening of the leadpipe, with variable R_2, is particularly important on the intonation of the instrument.

5. RESULTS

A. MACHINE LEARNING MODELS

For the spectral enrichment rate *m*, the comparison of models - each assessed on the training set via cross-validation under its optimal hyperparameter configuration - identified the Random Forest Regressor (RF) as the most effective. It achieved the lowest cross validated RMSE (0.0084) and a solid R^2 score (0.72), narrowly outperforming other ensemble methods such as Bagging and XGBoost.

The predictive performance of this retained RF model was evaluated on the independent test set. As shown in Table 1, the model achieved an RMSE of 0.0091, indicating a low average deviation from the true values. Considering that the original values of *m* range from approximately 0.086 to 0.175, this level of error is relatively small.

Table 1. Performance metrics of the best predictive model for m on the test set.

Model	RMSE	R^2	MAPE (%)
Random forest (RF)	0.0091	0.6692	5.30

The coefficient of determination, $R^2 = 0.6692$, indicates that approximately 67% of the variability in *m* is accounted for by the model. These results are satisfactory, though alternative approaches could be considered to further improve performance. Additionally, the Mean Absolute Percentage Error (MAPE) of 5.30% indicates that, on average, the model's predictions deviate by 5.3% from the true values. This represents a reasonable level of accuracy for regression tasks of this nature.

For the average normalized spectral centroid $\overline{NSC}$, the comparison of models identified the XGBoost regressor (XGB) as the top performer. It achieved the lowest cross validated RMSE (0.0340) and the highest R^2 score (0.92), slightly outperforming other ensemble methods such as RF and Bagging.

The model's predictive performance on the unseen test set is summarized in Table 2.

Table 2. Performance metrics of the best predictive model for $\overline{NSC}$ on the test set.

Model	RMSE	R^2	MAPE (%)
XGBoost	0.0310	0.9047	0.76

The model exhibited excellent predictive performance for the descriptor $\overline{\mathrm{NSC}}$.

To verify the accuracy of the adopted prediction models for both descriptors, a subset of 8 trumpets from the test set - unseen during model training - is presented in Fig. 5, comparing their predicted positions in the descriptor space with those obtained from simulations using the physical model. These results suggest that the models under consideration show reliable predictive performances, with most trumpet samples showing a close match between simulated and predicted descriptor values.

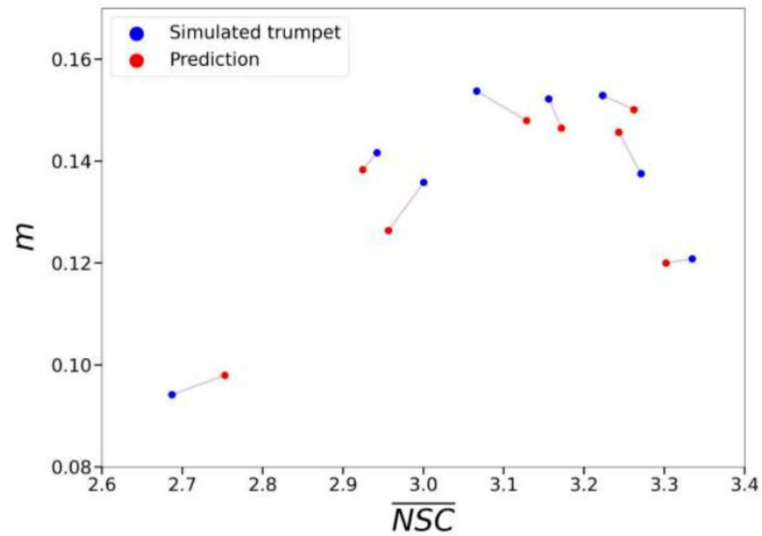

Figure 5. Prediction error for 8 trumpets excluded from training

B. PREDICTIONS OF THE DESCRIPTORS

To sample the design space, 12 equally spaced values are selected for each parameter from a grid discretization defined between its lower and upper bounds in the original dataset. The full mesh of possible combinations—consisting of 12^6 = 2,985,984 in total—is then filtered to retain only those satisfying the increasing radii condition. This results in approximately 306,000 valid leadpipe geometries.

Given that ML models of EFP were defined in previous studies[10] on the design space, we are able to incorporate an intonation constraint in the analysis, allowing us to focus only on instruments with satisfactory tuning. Thus, we retain only those with a low predicted pitch deviation - specifically, EFP < 15 cents - ensuring reasonably accurate intonation, as lower values indicate better tuning. This allows us to examine how the predicted timbre descriptors are distributed among the instruments that satisfy the intonation constraint.

After applying the intonation-based selection, the positions of the trumpet samples according to the ML models for the two descriptors - *m* and $\overline{\mathrm{NSC}}$ are shown in Fig. 6. The prediction values for $\overline{\mathrm{NSC}}$ ranges from 3 to 3.3, and *m* values approximately between 0.11 and 0.16.

In addition, predictions are also carried out for two leadpipes of particular interest in this study:

- The standard leadpipe of a commercial, denoted “S”,
- The “NL” leadpipe, derived from previous works (Ref. 11).

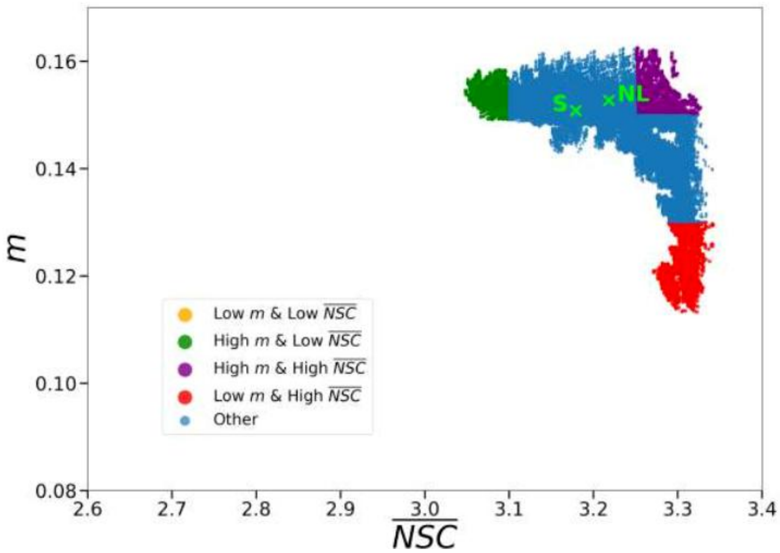

Figure 6. Biplot of the predicted positions of the trumpets in the descriptor space obtained by the ML predictions, for instruments with EFP < 15 cents

As illustrated in Fig.6, the trumpet geometries can be classified into three categories based on the limit values of the two descriptors:

- High m and Low $\overline{\text{NSC}}$ — defined by ($\overline{\text{NSC}}$ <3.1, m>0.15), shown in green. These trumpets exhibit significant spectral centroid variation in response to pressure, despite having low $\overline{\text{NSC}}$ values,
- High m and High $\overline{\text{NSC}}$ — defined by ($\overline{\text{NSC}}$ >3.25, m>0.15), represented in purple. Instruments in this category combine a high $\overline{\text{NSC}}$ with strong spectral centroid growth,
- Low m and High $\overline{\text{NSC}}$ — defined by ($\overline{\text{NSC}}$ >3.2, m<0.13), depicted in red. These trumpets display high $\overline{\text{NSC}}$ values but exhibit little increase of brightness with pressure

Once the trumpets are categorized, we examine the distribution of geometries within each class.

C. VIOLIN PLOTS

Fig. 7 illustrates the distribution of leadpipe geometries across categories, highlighting the influence of the general leadpipe shape on the sound descriptors.

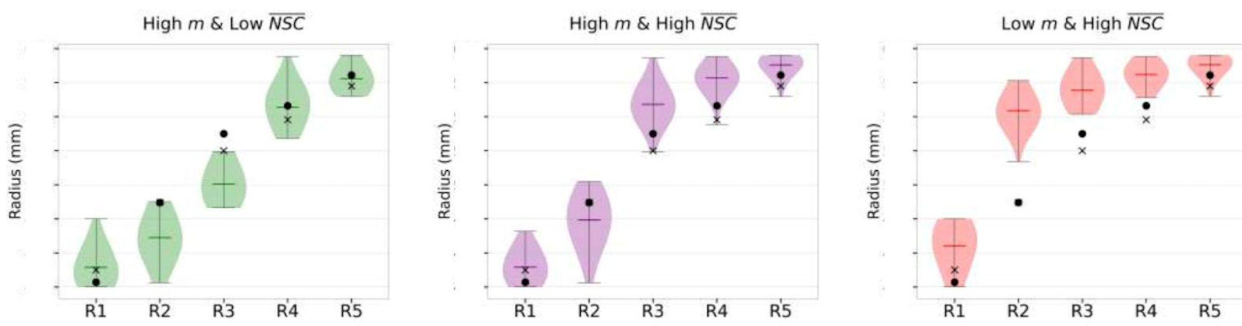

Figure 7. Comparative distribution of geometries within the 3 defined categories. For confidentiality reasons, radii values have been removed

The "High m and Low $\overline{\text{NSC}}$" class displays a gradual increase across radii. Meanwhile, the "High m and High $\overline{\text{NSC}}$" category is associated with a pronounced rise between R2 and R3, followed by stabilization with smaller incremental increases in subsequent radii. The "Low m and High $\overline{\text{NSC}}$" class, by contrast, exhibits an abrupt increase between R1and R2, which then stabilizes.

Additionally, the influence of radii R1 and R2 on the spectral profiles of the instruments is clearly observable. Specifically, for the categories "High m and Low $\overline{\text{NSC}}$", and "High m and High $\overline{\text{NSC}}$", the values of R1 and R2 are relatively similar. In contrast, the "Low m and High $\overline{\text{NSC}}$" category exhibits a significant increase between the two radii. Moreover, among the defined classes, the geometries of the NL and S trumpets closely resemble the profile of the "High m and High $\overline{\mathrm{NSC}}$" category, as evidenced by Fig. 6.

Finally, both the particular distribution of geometries across categories and the predicted behavior of the "NL" and "S" trumpets highlight the relationship between leadpipe geometry and the considered descriptors. Consequently, this offers a preliminary guideline to steer the design of the leadpipe according to the desired instrument profile in terms of the two proposed descriptors.

By and large, the study reveals three main leadpipe geometries:

- To reach low $\overline{\mathrm{NSC}}$ with high m, a progressive opening from R1 to R3 is necessary, with R1 and R2 remaining narrow and close to each other,
- To achieve both high $\overline{\mathrm{NSC}}$ and high m, R1 should be small, and R2 should remain moderate.
- To obtain high $\overline{\mathrm{NSC}}$ with low m, a significant difference between R2 and R1 is required.

Variables R4, R5 do not play a discriminant role in separating the categories. In conclusion, R1, R2 and R3 emerge as the most influential parameters for distinguishing the categories under study. Experimental validation with real trumpets would be necessary to confirm these findings.

6. CONCLUSION

The methodology presented in this paper shows how sound simulations and Machine Learning models can be used to study the sound qualities of brass instruments. It was applied to the trumpet and to two descriptors calculated from the spectral centroid of sounds: $\overline{\mathrm{NSC}}$, the average normalized spectral centroid of the instrument across different blowing pressures, and *m* the slope of the spectral enrichment, representing the rate of increase of the normalized spectral centroid with respect to blowing pressures.

Violin plots proved useful in highlighting distribution patterns, while bi-optimization and multi-criteria decision analysis enabled the selection of precise shapes (not presented here). The study aimed to explore new relationships between leadpipe geometry and spectral descriptors, and to identify shape patterns potentially associated with specific acoustic behaviors. Typical geometries of leadpipes, provided by this study and made in rapid prototyping, are currently being tested in order to confirm the validity of the approach.

Different perspectives can be drawn. Non-linear propagations in the resonator should be included to get realistic timber predictions. It will be also very interesting to include other spectro-temporal descriptors, or transient regimes.

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